

1 Cloud Computing Fundamentals

Cloud Computing

Has characteristics



- On-Demand self service
- Broad Network Access
- Resource Pooling
- Rapid Elasticity
- Measured service

It is defined as On-demand delivery of computing resources over the Internet with pay-as-you-go model



Provides Benefits.

Cost Savings

Scalability & Elasticity

Flexibility

High Availability

Focus on Core Business

Has limitations



- Security & privacy concerns
- Dependence on Internet
- Limited control & customization
- Compliance challenges
- Vendor lock-in

2. EVOLUTION OF CLOUD COMPUTING.

Evolution of cloud computing

Evolved from

1. Hardware Evolution

- Mainframes
- Client-server architecture
- Virtualized servers
- Multi-core processors
- Commodity hardware

2. Internet Software Evolution

- Internet growth
- Web Technology (HTML, HTTP)
- Web applications
- Service-oriented architecture
- APIs & Mashups

3. Distributed Systems Evolution

- Cluster computing
- Grid computing
- Utility computing
- Autonomic computing
- Resource Management

4. Service Delivery Evolution

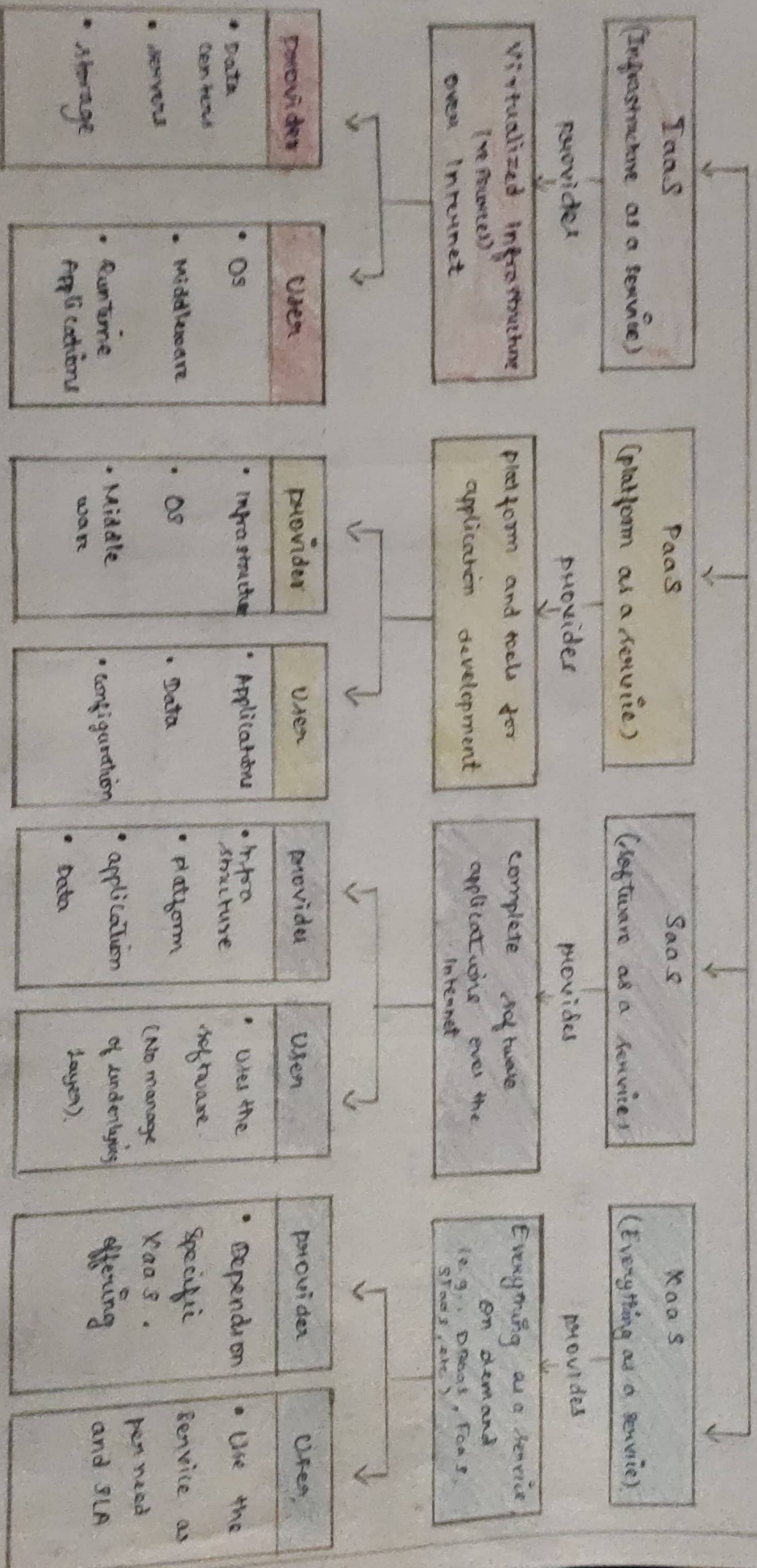
- On-demand provision
- Pay-as-you-go
- Multi-tenancy
- SLA & QoS
- Cloud Services (IaaS, PaaS, SaaS)

CLOUD COMPUTING

On-demand, Scalable, Measured services.

3. CLOUD SERVICE MODELS

Cloud Service Models.



4. VIRTUALIZATION & WEB SERVICES AS ENABLERS OF CLOUD

